Undocumented population estimates

A reduced-form implementation of the Warren, Warren & Zheng (2023) residual method

Adjustment: ACS noncitizens with Pew, MPI, and CMS anchors
United States, Texas, and Texas health-insurance status, 2008–2024

Texas 2036 Data & Research · Draft · July 2, 2026

Method: Warren residual method, applied to ACS

Warren, Warren & Zheng (2023):

Undocumented = noncitizens − legal noncitizens:

$$U_t = C_t - L_t$$

Warren rolls L_t forward each year from DHS lawful admissions, naturalizations, life-table deaths, and (their key innovation) emigration derived from ACS movement.

This workflow:

C_t comes straight from ACS table B05001. θ_t is the **undocumented share of noncitizens** ($\theta_t = U_t/C_t$). Because legal noncitizens change slowly, θ_t moves slowly too.

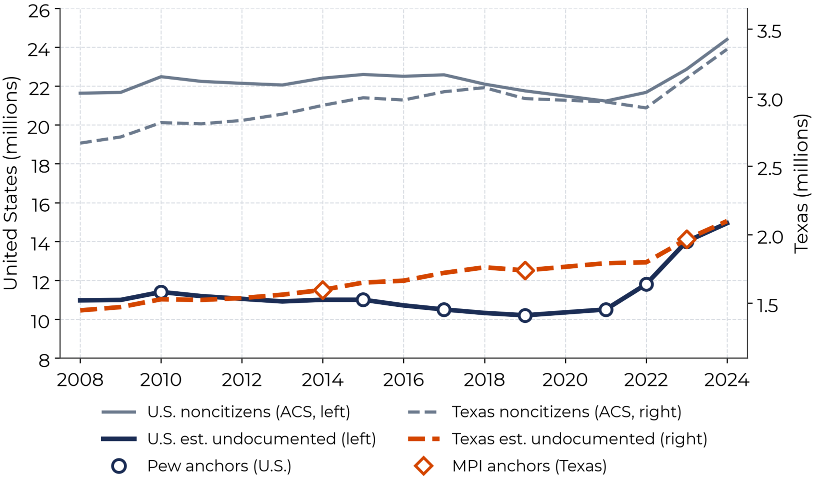
So we pin θ_t to the values Pew, MPI, or CMS publish at their anchor years, and interpolate between. The estimate reproduces each source's published number exactly at anchor years.

Not an "ACS fill-in" of a fixed number

Two versions of this same residual method presented in parallel:
one uses Pew (U.S.) and MPI (Texas) anchors; the other uses Warren/CMS anchors.
Both are residual estimates, not a benchmark check of the other. Each inherits its source's demographic accounting (admissions, deaths, emigration, undercount) through the anchor value.

Source: Texas 2036 analysis of ACS via Stata `getcensus`; Warren, Warren & Zheng, *International Migration Review* (2023), DOI 10.1177/01979183231195280.

2024 estimate: 14.9M nationally and 2.10M in Texas



U.S., 2024 (Pew/MPI vintage)

14.9M

4.4% of total U.S. population

Texas, 2024 (Pew/MPI vintage)

2.10M

6.7% of total Texas population

Texas, 2024 (Warren/CMS)

2.32M

7.4% of total Texas population

Markers are published residual-method values (Pew U.S., MPI Texas); $\widehat{U}_t = \theta_t \times C_t$ passes through them exactly and interpolates θ elsewhere.

Texas count rises in 2024 while the uninsured rate falls

Year	ACS noncitizens (millions)	Pew/MPI estimate (millions)	Undoc. share of TX pop.	Warren/CMS check (millions)	Insured (millions)	Uninsured (millions)	Uninsured rate among undoc.
2010	2.82	1.53	6.1%	1.72	0.32	1.20	79%
2015	3.00	1.65	6.0%	1.76	0.51	1.14	69%
2019	2.99	1.74	6.0%	1.78	0.54	1.20	69%
2021	2.97	1.79	6.1%	1.85	0.57	1.23	68%
2022	2.92	1.80	6.0%	1.87	0.59	1.21	67%
2023	3.14	1.97	6.4%	2.05	0.66	1.31	66%
2024	3.35	2.10	6.7%	2.32	0.76	1.34	64%

notes: ACS noncitizens is the raw, observed count from the Census table, incl. every non-citizen in Texas: green-card holders, student/work-visa holders, refugees, asylees, and undocumented people.
The Pew/MPI residual-method vintage estimates the undocumented subset of that column by pinning $\theta_t = U_t / C_t$ to MPI's published Texas value at anchor years.

2024 Pew/MPI vintage

2.10M

estimated undocumented Texans

2024 uninsured

1.34M

64% of estimated undocumented Texans

Pew/MPI vintage pins θ_t to MPI's published Texas values in 2014, 2019, and 2023; other years interpolate θ, with ACS C_t supplying the count movement. The Warren/CMS column pins to Warren/CMS state anchors for 2010–2019, 2023, and 2024.

Source: Texas 2036 analysis of ACS B05001 and B27020; MPI Texas profiles (2019, 2023); CMS State-by-Year 2010–2019 dataset (annual 2010–2019) and Warren/CMS data tool (2023–2024). Pew/MPI Texas adds a pooled 2012–16 MPI anchor at 2014; 2008–2013 extrapolate it.

Three residual-method sources agree on 2023 within a narrow band

Source (2023)	U.S.	Texas
Warren/CMS	12.24M	2.05M
Pew	14.00M	2.05M
MPI	13.74M	1.97M
This workflow (Pew/MPI vintage)	14.00M	1.97M

- All three published sources use variants of the Warren residual method ($U = C - L$). 2023 spread across them: about 14% nationally and 4% in Texas.
- Our Pew/MPI Texas 2023 value equals MPI's 1.97M by construction — about 4% below Warren/CMS and Pew, at the low end of the residual-method range, not outside it.
- In 2024, the Pew/MPI vintage lands at 14.94M nationally, about 2% above Warren/CMS's directly published 14.61M — two independent residual constructions converge.

Source: Immigration Research Initiative 50-state comparison (Nov 2025); Warren/CMS 2024 State and National Data Tool; Texas 2036 workflow output `triangulation_2023.csv`.

Why the shortcut works: the undocumented share of noncitizens barely moves

The share of noncitizens who are undocumented, θ, is a slow-moving number. It shifts only as the legal noncitizen population turns over, through small yearly flows of admissions, naturalizations, deaths, and departures. Nationally it drifts from about **0.52 in 2010 to 0.60 in 2023**, roughly a point a year. A number that steady can be pinned at a few published years and interpolated in between, with the annual ACS carrying the rest. That is the whole method.

Estimate at any year

$$\widehat{U}_{g,t} = \theta_{g,t} \times C_{g,t}$$

$$\theta_{g,a} = \frac{U_{g,a}^{\text{published}}}{C_{g,a}} \text{ at anchors } a, \text{ interp. between}$$

Two versions, same method

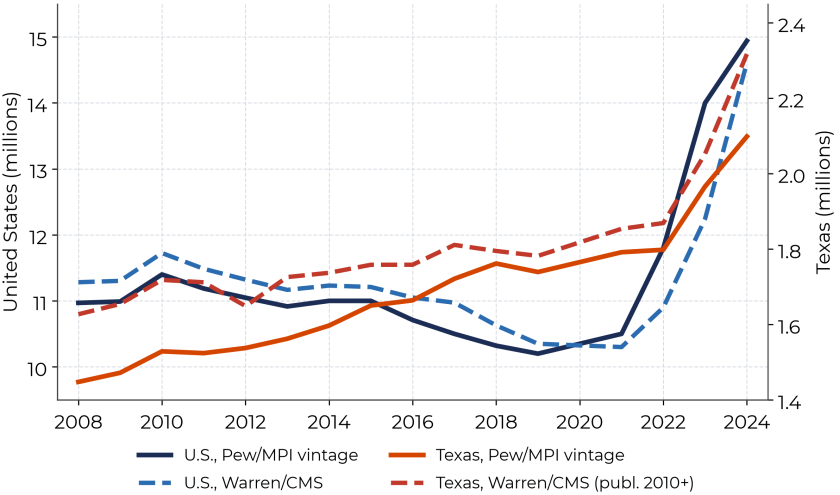
Pew/MPI: Pew U.S. 2010–2023 (7 anchor years), MPI Texas 2019 and 2023

Warren/CMS: Warren U.S. 2010–2024, Warren/CMS Texas 2010–2019, 2023, 2024

What we borrow, and what we don't

At each anchor year the estimate **equals the source's published number exactly**; the ACS noncitizen count fills the years between. So we inherit the hard demographic accounting — DHS admissions, life tables, emigration, undercount — straight from Pew, MPI, and CMS, instead of rebuilding it. The one stretch that is not a published number is the Texas line before 2019, which we extend back from the 2019 MPI anchor.

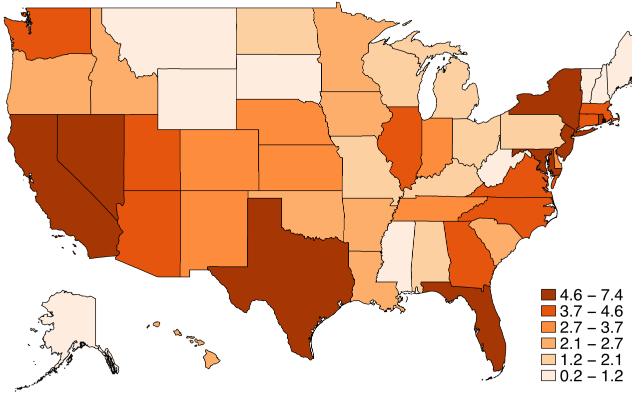
Two versions of the same residual method: diverge in the 2010s, converge in 2024



- Both lines use the same ACS C_t; they differ only in whose published U is used at anchor years.
- Warren/CMS ran higher in the 2010s; Pew's 2022–23 revisions flip that in 2023.
- In 2024 the two independent constructions land within 2% of each other.

Undocumented residents are 4.3 percent of the U.S., but 6–7 percent of Texas, California, and Florida

Undocumented residents as a share of state population, 2024 (%)



CMS direct state residual estimates over ACS 2024 population.

Undocumented, United States, 2024

14.6M

4.3% of the U.S. population

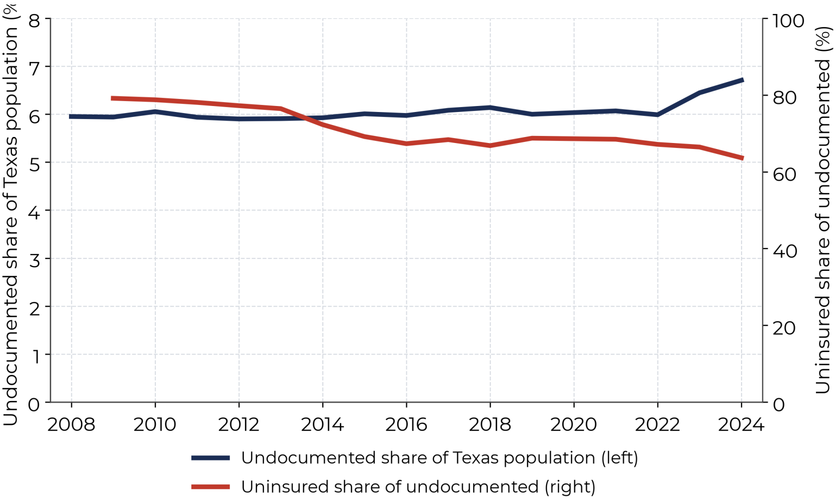
Texas, the highest large-state share

7.4%

tied with NJ; CA 6.7%, FL 6.1%

- The population is concentrated: the 12 largest states hold about three-quarters of the national total.
- Shares are highest in Texas and New Jersey (7.4%), then California, Florida, and Nevada, and lowest across the Upper Midwest and northern New England.
- The same reduced-form method that tracks Texas over time (previous slides) produces every state's 2024 level from one national anchor and each state's ACS noncitizen count.

Texas undocumented share reaches 6.7 percent



Undocumented share

6.7%

of all Texans in 2024

Uninsured share

64%

of estimated undocumented Texans

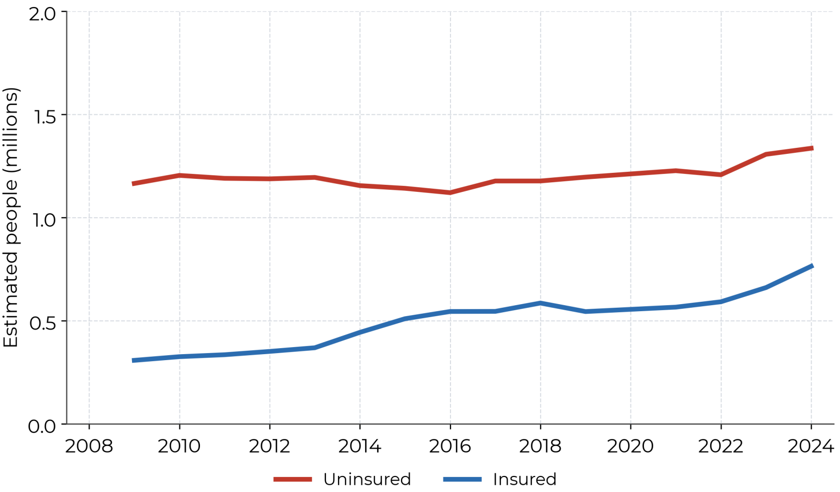
Insured share

36%

of estimated undocumented Texans

Chart shows the uninsured share only; the insured share is its complement.

Most estimated undocumented Texans lack health insurance



Uninsured, Pew/MPI vintage

1.34M

64% of estimated undocumented Texans

Insured, Pew/MPI vintage

0.76M

36% of estimated undocumented Texans

Uninsured undoc.

4.3%

of all Texans in 2024

PUMS is not needed for the top-line Texas insurance count

What the reduced-form residual estimates here

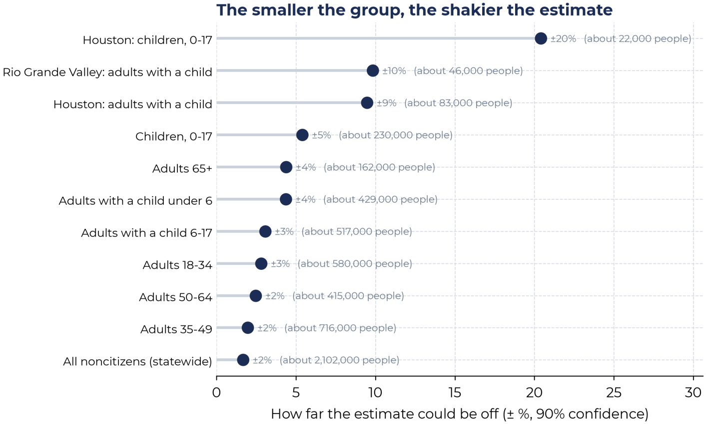
- Aggregate Texas undocumented insured and uninsured counts.
- ACS table B27020 already reports Texas noncitizen insurance totals; those are our observable C_t^{ins}.
- MPI's residual-method Texas 2023 uninsured rate anchors the odds-scale rate transport for θ_t^{ins}.

When PUMS becomes necessary

- Subgroup rates by age, origin, year of entry, income, education, or smaller geography.
- Person-level likely-legal-status imputation (e.g., an MPI/SIPP-style assignment for those doing the full Warren residual directly).
- Cross-tabs not available in published ACS tables.

Source: ACS B27020 provides aggregate noncitizen insurance counts; ACS PUMS is the person-level microdata file needed for subgroup modeling and for a direct year-by-year Warren reconstruction.

PUMS unlocks subgroups, but the margin of error grows as the group shrinks



Statewide undocumented Texans

±2%

90% sampling margin (about 2.1M)

Undoc. children in one county

±20%

90% sampling margin (about 22,000)

- Same method, finer cells: apply the state θ and insurance transport to PUMS subgroups. These are transported estimates, not person-level status imputations.
- Detail is real and useful for targeting: families with young children, or the Rio Grande Valley at 76% uninsured.
- **So what**: read small-cell numbers as ranges, not points. The margin shown is ACS sampling error; cross-source and undercount uncertainty add to that.

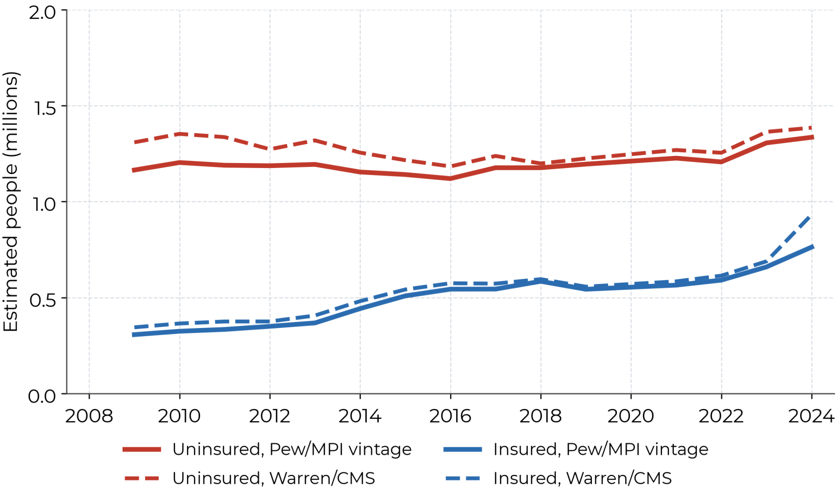
How do we get an insurance rate for people the survey can’t identify?

No survey record is labeled “undocumented,” so we cannot read their uninsured rate directly. We build it from a rate we *can* see.

- **What we can see:** the ACS reports the uninsured rate for *all* noncitizens together — green-card holders, temporary legal residents, and the undocumented.
- **The key fact:** the undocumented are uninsured at a *higher* rate than legal noncitizens (barred from most public coverage, fewer job-based plans), so the combined rate is a floor.
- **The move:** borrow one published undocumented rate (MPI for Texas, CMS nationally), measure how far it sits above the noncitizen rate, apply that same lift wherever we have the noncitizen rate (on the odds scale, so it never exceeds 100%), and multiply by the undocumented count.
- **Honesty:** where CMS publishes the split directly (states, 2024) we use it; otherwise it rests on one benchmark, so we show a band (Texas 60–66%; about 45% nationally, where KFF and CMS agree), not a single line.

Source: ACS table B27020 (noncitizen insurance); MPI and CMS undocumented insurance benchmarks; KFF coverage-by-status.

CMS puts Texas uninsured undocumented at 1.39M



Pew/MPI vintage uninsured

1.34M

2024 estimate (64% uninsured)

Warren/CMS uninsured

1.39M

2024 published value

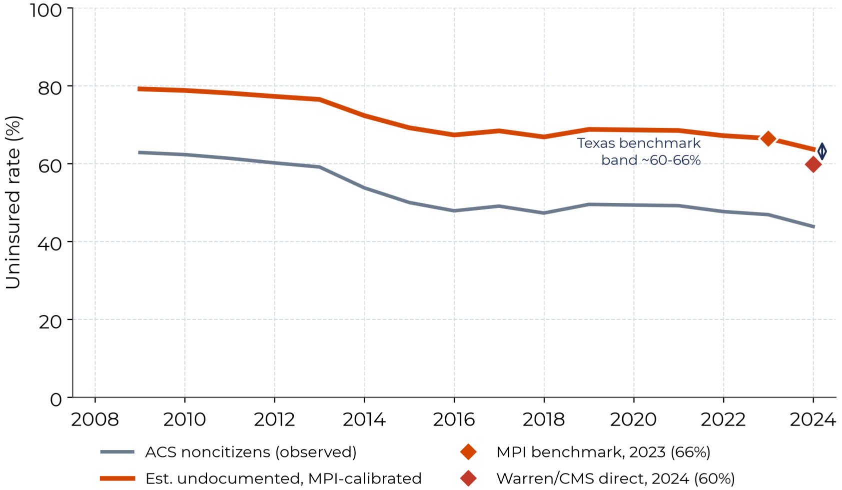
Warren/CMS uninsured rate

60%

of Warren/CMS estimated undocumented Texans

Warren/CMS's 2024 Texas value is a directly published residual estimate, independent of the Pew/MPI vintage's ACS/MPI rate transport — two

Estimated undocumented Texans are far more likely to be uninsured



Estimated undocumented

64%

uninsured in 2024

ACS noncitizens

44%

uninsured in 2024

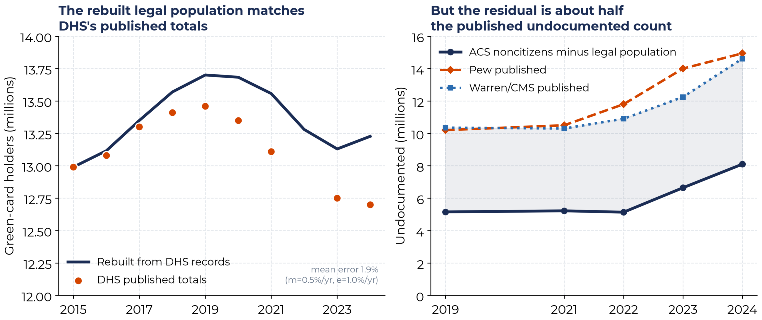
MPI 2023 (66%) and Warren/CMS direct 2024 (60%) bracket the rate as a 60–66% band; nationally KFF and CMS agree near 45%. Pre-2014 rates are more uncertain.

How to read these numbers

- **Bounded estimates, not direct counts.** Each vintage exactly reproduces its source's published U_t at anchor years; ACS C_t carries between-anchor movement.
- **Two tails extrapolate** the Pew/MPI vintage — U.S. 2008–09 and Texas 2008–13 — so those early Texas years are indicative, not published.
- **Method break at 2022–23.** Pew and Warren/CMS both revised their residual methods in that window; some of the sharp rise is definitional, not demographic.
- **Warren/CMS Texas 2010–2024 is published Warren/CMS data.** Only 2008–09 extrapolate. It already carries admissions, mortality, ACS-derived emigration, and undercount adjustments.
- **"Insurance" means undocumented coverage, not noncitizen coverage.** ACS (B27020) reports the noncitizen uninsured rate for every state; the undocumented subset is not observed, so it needs a benchmark. The Pew/MPI vintage calibrates to MPI's 2023 Texas rate, so its insurance line is Texas-only. Warren/CMS's 2024 release directly publishes undocumented insurance nationally (45% uninsured, matched by KFF's 45% for 2017) and by state; for Texas we show a 60–66% band (MPI-calibrated to CMS-direct) rather than one line. Insurance-rate uncertainty is not fully propagated.

Source: Estimation code: `code/01_acs_residual_undocumented_estimates.do`; Warren, Warren & Zheng, *International Migration Review* (2023).

Can we rebuild the numbers from raw government data?



Legal residents: yes. Green-card admissions, minus naturalizations, deaths, and departures, reproduce the government's published green-card totals within 2% (left chart).

Undocumented: not from scratch. Subtracting the legal population from survey noncitizens gives only about half the published total (right chart), because government records drop people who leave or die more slowly than the survey does. Pew and CMS already correct for this, so we anchor to them.

Source: DHS OHSS Yearbook and population estimates; ACS noncitizens; Pew and Warren/CMS published undocumented counts.

Where the method stands, and what's next

In place: each vintage anchors on its own multi-year source (Pew U.S. 2010–2023; MPI Texas 2014/2019/2023; Warren/CMS U.S. 2010–2024, Texas annual 2010–2019 plus 2023–2024); a worked Texas PUMS disaggregation to decision-relevant subgroups; uncertainty reported two ways (an ACS sampling margin plus a separate cross-source range); an insurance band (Texas 60–66%, MPI-calibrated to CMS-direct; KFF and CMS near 45% nationally); and a from-scratch build that reproduces the government's published legal-population totals within 2%.

What's next:

- Keep refining the from-scratch build so it can reproduce the full undocumented count directly, not just the legal population. For now the published-ratio method stays the practical tool, because it already carries the coverage adjustment a from-scratch build has to estimate.

Source: Warren, Warren, and Zheng, *International Migration Review* (2023), DOI 10.1177/01979183231195280; DHS OHSS; Pew, MPI, and CMS residual-method documentation.

Sources and reproducibility

- ACS 1-year detailed tables B05001 (2008–2024) and B27020 (2009–2024), pulled with Stata `getcensus`; 2020 standard ACS 1-year unavailable.
- Pew Research Center, *U.S. Unauthorized Immigrant Population Reached a Record 14 Million in 2023*, August 21, 2025; and the accompanying methodology Q&A (August 22, 2025), which revised 2022 to 11.8 million. Published historical values used as validation markers.
- Migration Policy Institute, *Profile of the Unauthorized Population: Texas* (1,966,000; 1,306,000 uninsured), and *Changing Origins, Rising Numbers* fact sheet (13.7M national); values verified on the live MPI pages July 2, 2026.
- Immigration Research Initiative, *50 States: Immigrants by Number and Share*, November 10, 2025 (side-by-side CMS/Pew/MPI 2023 state estimates).
- Center for Migration Studies of New York, *Understanding the U.S. Undocumented Population: New 2024 Estimates from CMS*, May 5, 2026; State and National Data Tool CSV accessed July 2, 2026.
- Warren, Warren, and Zheng, "A New Residual Approach for Estimating Undocumented Populations," *International Migration Review*.
- Reproducible code: `code/01_acs_residual_undocumented_estimates.do` (canonical) and `code/02_rebuild_outputs_python.py` (mirror); assumptions log in `data/processed/method_assumptions.csv`.